Statistical Analysis Report:

Understanding the Modern Traveller: A Market Analysis for a New Travel Agency &

Illicit Drug Use Among Young People: Policy Guidelines

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Understanding the Modern Traveller: A Market Analysis for a New Travel Agency

Exploratory Data Analysis

Purpose: To summarize the key features of the 250 respondents and illustrate insights from the samples before executing inferential testing.

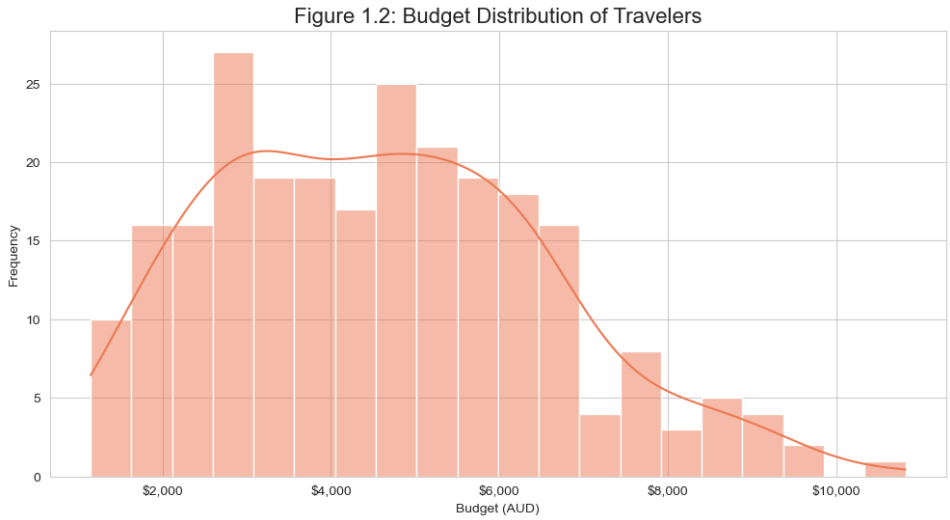
Numerical Variables

```
--- Table 1.1: Descriptive Statistics for Numerical Columns ---
       Q1_Age  Q8_Budget
count  250.00     250.00
mean    43.85    4593.08
std     13.25    1974.20
min     20.00    1148.00
25%     33.00    2995.50
50%     44.00    4544.00
75%     53.75    5926.25
max     75.00   10823.00
```

Age Distribution of Travelers

According to table 1.1 and figure 1.1, the age distribution of the 250 respondents appears roughly symmetrical and quite normally distributed, centered around the mean age of 43.85 years. This falls within the middle-aged adult range. The standard deviation is 15.6 years, indicating a fairly wide spread of ages in the sample, ranging from 20 to 75 age bracket but with significant representation across most adult age groups.

Budget Distribution of Travelers



The average travel budget per person for a one-week trip is $4,593.08 AUD. The distribution is slightly right-skewed, as indicated by the mean being slightly higher than the median ($4,544 AUD). This means that while most travelers budget within a certain range, a smaller number of individuals have significantly higher budgets, pulling the average up. Figure 1.2 illustrates this, showing a concentration of budgets between approximately $2,000 and $6,000, but with a tail extending towards higher values. The standard deviation is substantial ($1,974.20 AUD), reflecting considerable variability in spending plans among travelers.

Key Categorical Variables

While the dataset contains numerous categorical variables, this section focuses on the most relevant factors for understanding traveler demographics, behaviors, and preferences, which directly inform the business strategy for the new travel agency.

Distribution of Marital Status

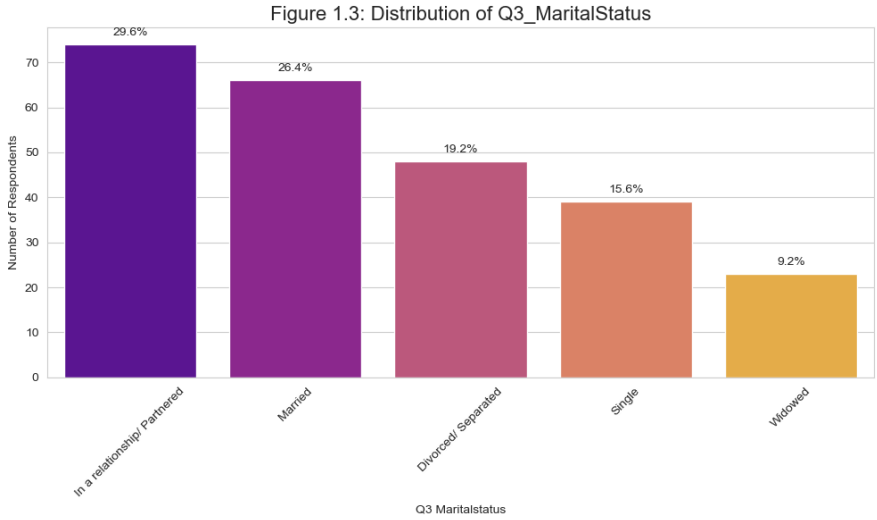


Figure 1.3 shows that the top largest group of respondents are 'In a relationship/ Partnered' and 'Married', followed by those who are 'Divorced/ Separated' and 'Single', suggests the agency's challenging areas are in solo-travelers side.

Distribution of Employment Status

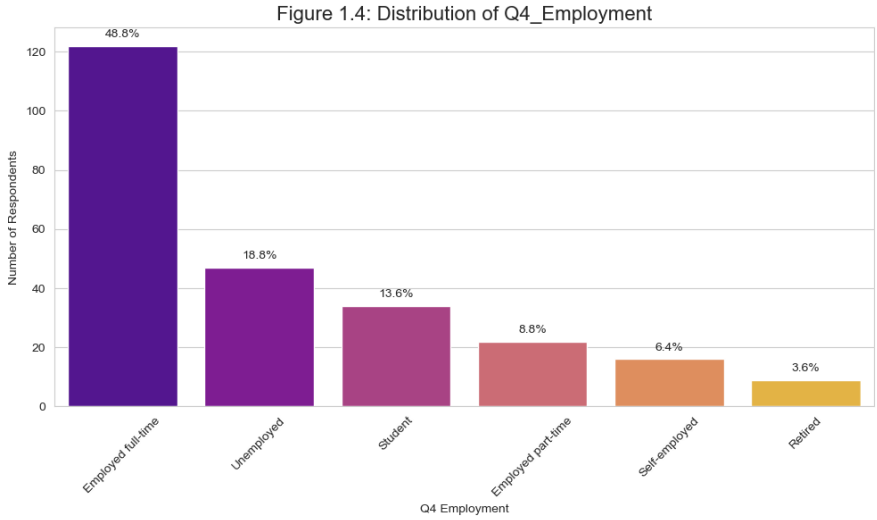
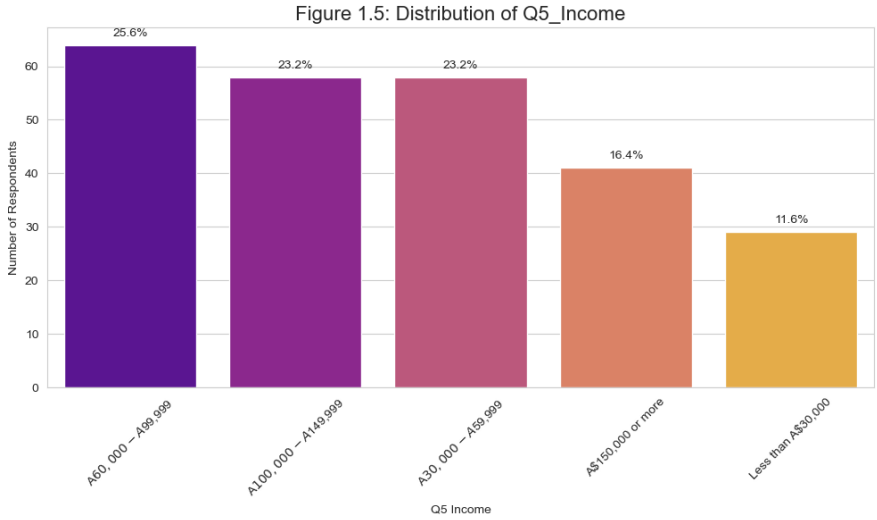


Figure 1.4 illustrates that the majority of the sample is 'Employed full-time', indicating a substantial market segment with stable income potentially available for travel.

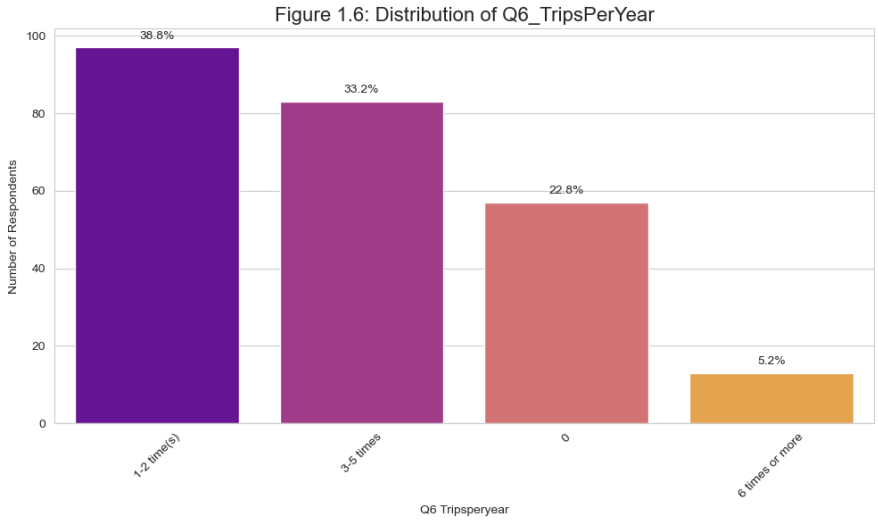
Distribution of Estimated Annual Household Income (AUD)



Note: Q5_Income represents estimated annual household income, not individual income.

The household income distribution is relatively spread out, but the largest concentrations appear in the middle to upper-middle income brackets (e.g., $60,000-$149,999 AUD). This aligns with the average budget observed and suggests the target market generally has reasonable disposable income for travel.

Distribution of Leisure Trips Per Year



According to figure 1.6, most respondents travel 1-2 times per year, indicating that international travel is a significant but not extremely frequent activity for the average person in this sample.

Distribution of Holiday Types

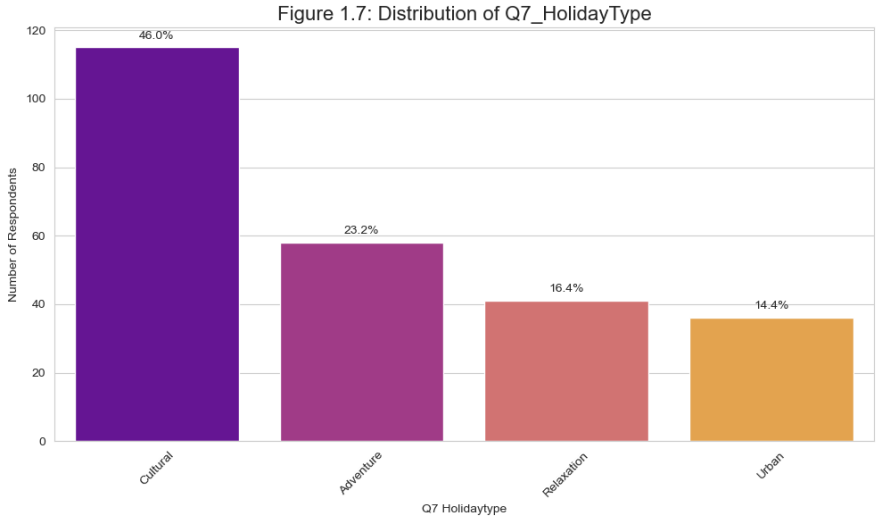
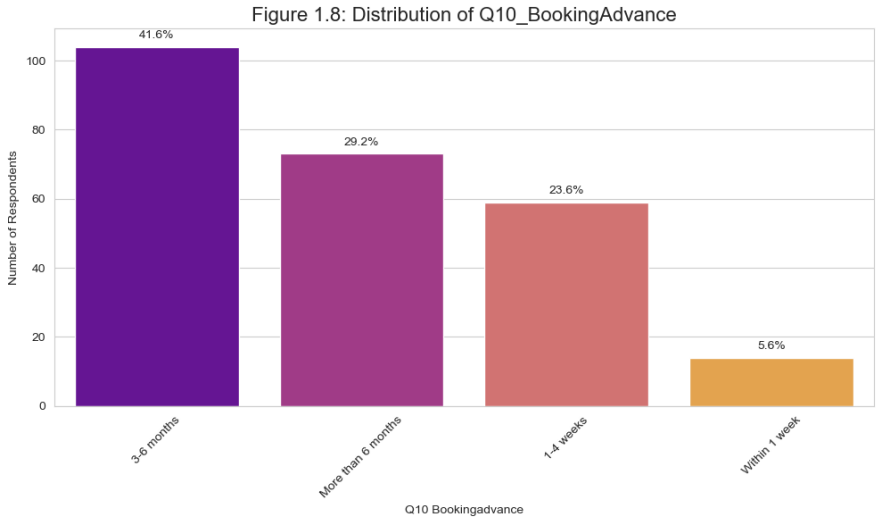


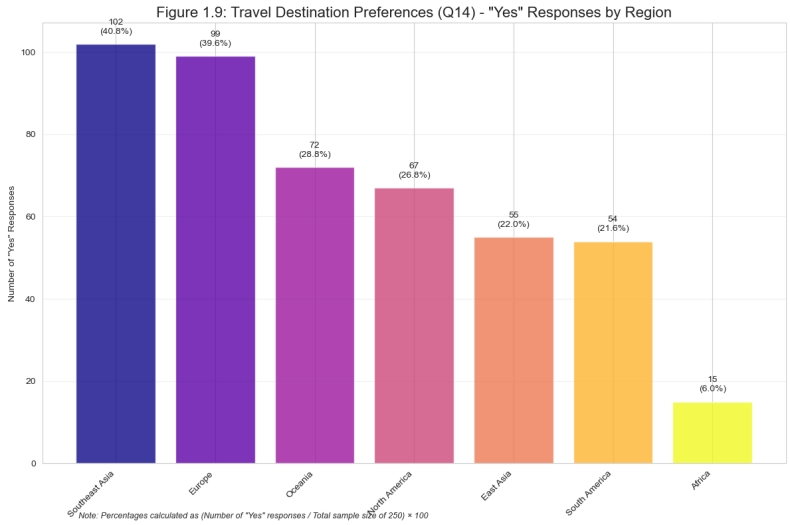
Figure 1.7 points out that 'Cultural' and 'Adventure' are the most popular holiday types, followed by 'Relaxation' trip. This highlights the need for diverse package offerings.

Distribution of Booking-Advance Preference



From figure 1.8, travelers show varied booking behaviors, but planning 3-6 months in advances seems most common, suggesting a window for targeted marketing campaigns.

Distribution of Travel Destination Preferences



'Southeast Asia' and 'Europe' are the most desired destinations by a significant margin, followed by 'Oceania' and 'North America'. These destinations can guide the direction for initial package development.

Distribution of Ability to Customize a Travel Package

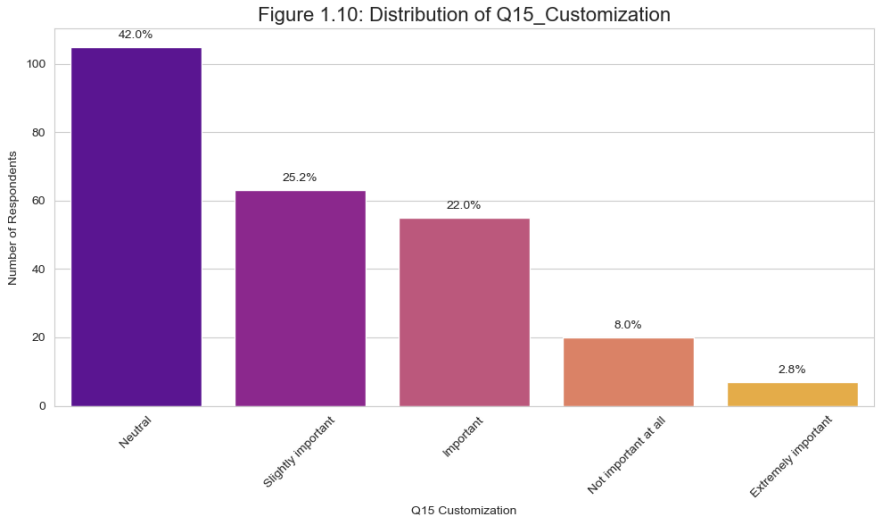
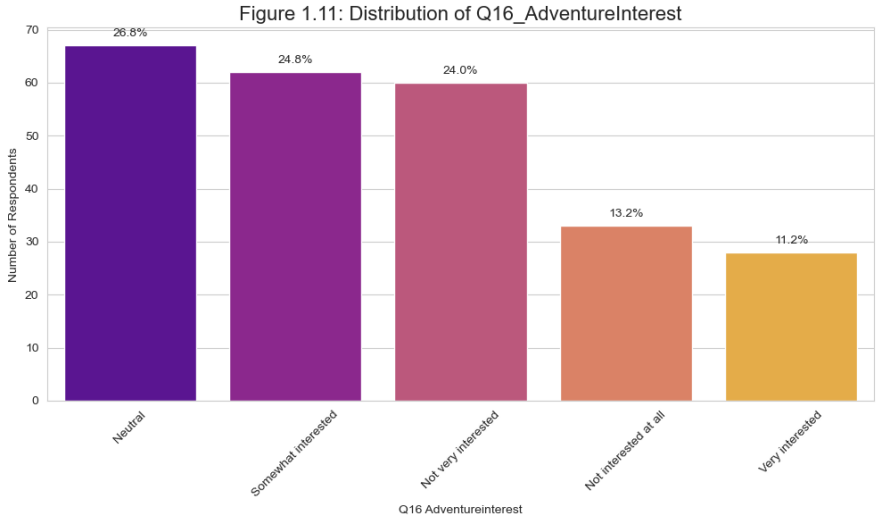


Figure 10 demonstrates that most travelers are neutral about customization of the package. Only some of them demand for flexible or personalized travel options, rather than purely fixed packages.

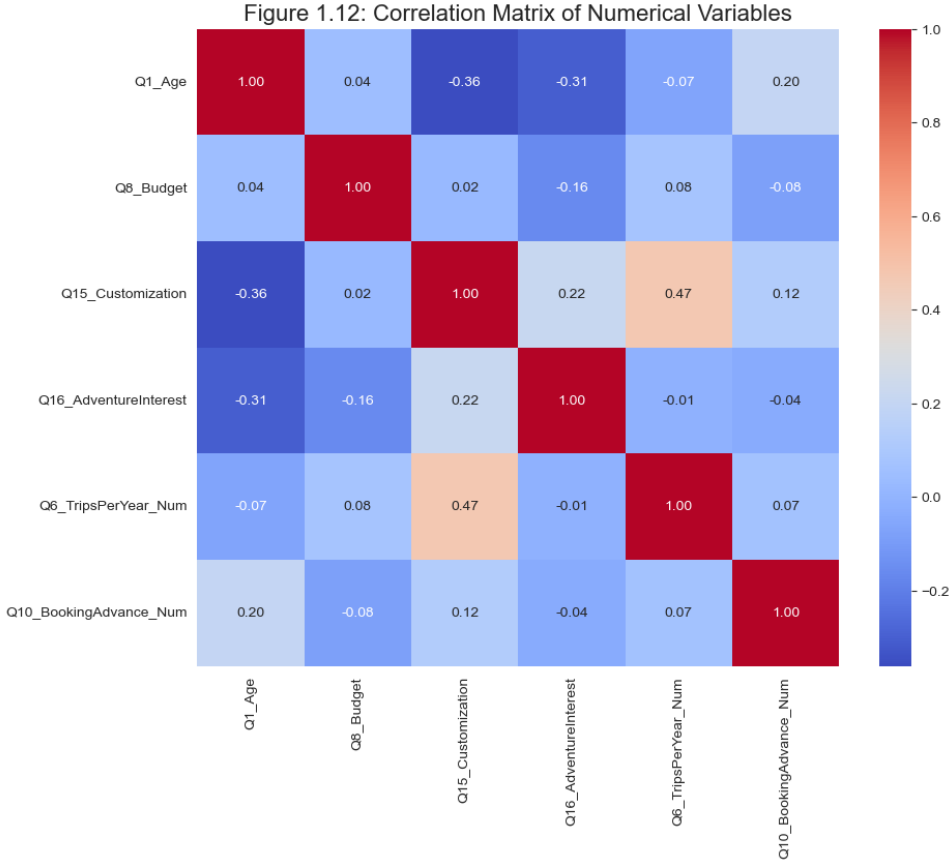
Distribution of Adventure-focused Interest



From figure 1.11, the interest in adventure is quite polarized. While a large group feel 'neutral' about it, 'somewhat interested' group and 'not very interested' group are almost equal.

Correlation Matrix of Numerical Variables

Figure 1.12: Correlation Matrix of Numerical Variables



The correlation matrix (Figure 1.12) reveals the linear relationships between the numerical variables in the dataset. Note that some variables included here, such as Customization Importance (Q15_Customization), Adventure Interest (Q16_AdventureInterest), Trips Per Year (Q6_TripsPerYear_Num), and Booking Advance (Q10_BookingAdvance_Num), are originally ordinal categorical variables that have been mapped to a numerical scale to enable this analysis. Red indicates a positive correlation, blue indicates a negative correlation, and the intensity signifies strength.

There is a moderate negative correlation between Age and both Customization (-0.36) and AdventureInterest (-0.31). This suggests that younger travelers tend to place a higher value on both trip customization and adventure compared to older travelers.

A strong positive correlation exists between the number of TripsPerYear and the importance placed on Customization (0.47), indicating that more frequent travelers tend to desire more personalized and tailored travel experiences. They are likely more experienced and have specific preferences, moving away from standard package tours.
Interestingly, there is almost no linear correlation between age and budget (0.04), suggesting that there is not a simple, straight-line increase in budget with every year of age. Budget is likely influenced more by factors such as household annual income and travel style within those age group.

Foundational Inferential Analysis

Purpose: To perform statistical tests to answer key business questions regarding traveler demographics and preferences.

95% Confidence Intervals of Traveler age

Purpose: To estimate the true population mean for traveler age and budget

```
--- Table 1.2: 95% Confidence Intervals for Travel Data ---
============================================================
                      Metric Value
                 Sample Size   250
            Mean Age (years) 43.85
95% CI for Age - Lower Bound 42.20
95% CI for Age - Upper Bound 45.50
============================================================
```

Table 1.2 indicates the mean age of the sample is 43.85 years. We can be 95% confident that the true mean age of the target traveler population lies between 42.20 and 45.50 years.

```
--- Table 1.3: 95% Confidence Intervals for Travel Data ---
============================================================
                         Metric     Value
                    Sample Size       250
              Mean Budget (AUD) $4,593.08
95% CI for Budget - Lower Bound $4,347.17
95% CI for Budget - Upper Bound $4,839.00
============================================================
```

Table 1.3 points out that the mean budget is $4,593.08 AUD. We are 95% confident that the true mean budget for a one-week trip in the target population is between $4,347.17 and $4,839.00 AUD.

ANOVA Test of Traveling Dataset

Purpose: To investigate whether the mean travel budget differs significantly across various traveler characteristics.

Table 1.4: ANOVA Results – Travel Budget by All Categorical Variables

Variable	F-Statistic	P-Value	Significant	Groups
Q5_Income	14.5043	0.0000	Yes	5
Q10_BookingAdvance	5.8054	0.0008	Yes	4
Q6_TripsPerYear	2.7759	0.0419	Yes	4
Q13_Accommodation	2.1844	0.0713	No	5
Q7_HolidayType	1.6983	0.1680	No	4
Q3_MaritalStatus	1.5712	0.1826	No	5
Q9_Priorities	1.1991	0.3039	No	8
Q11_InfoSources	1.1079	0.3583	No	7
Q19_FuturePriorities	0.8520	0.4667	No	4
Q4_Employment	0.9072	0.4770	No	6
Q18_TechTrends	0.7915	0.4996	No	4
Q12_BookingPreference	0.4029	0.6688	No	3
Q17_CustomServices	0.3433	0.7941	No	4

Table 1.4 presents the p-values from multiple ANOVA tests. Using significance level of alpha=0.05, we identified three categorical variables that shows a statistically significant relationship with the travel budget: Q5_Income, Q10_BookingAdvance, and Q6_TripsPerYear. Tests involving other variables, such as Q13_Accomodation (p=0.0713), Q7_HolidayType (p=0.1680), and the rest, were not found to be as strongly related or were not statistically significant and thus are omitted from further detailed analysis. The following sections delve deeper into the three significant findings.

ANOVA Results – Travel Budget by Household Income Status

Table 1.5: ANOVA Results – Travel Budget by Household Income Status

Descriptive Statistics by Group:

A$100,000–A$149,999	N= 58	Mean= 5207.12	SD= 2135.80	Range=1389.00–10823.00
A$150,000 or more	N= 41	Mean= 5893.78	SD= 1814.35	Range=2207.00– 9736.00
A$60,000–A$99,999	N= 64	Mean= 4524.25	SD= 1774.62	Range=1148.00– 9178.00
Less than A$30,000	N= 29	Mean= 3093.31	SD= 1072.51	Range=1231.00– 6638.00
A$30,000–A$59,999	N= 58	Mean= 3885.43	SD= 1687.71	Range=1282.00– 8135.00

ANOVA Table:

Source	SS	df	MS	F	p-value
Between Groups	185811013.63	4	46452753.41	14.5043	0.0000
Within Groups	784656591.61	245	3202679.97	–	–
Total	970467605.24	249	–	–	–

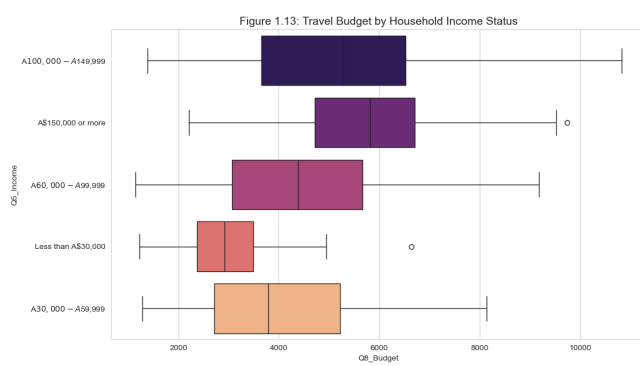


Table 1.5 confirms a highly statistically significant difference in travel budgets based on household income ($p < 0.0001$). As logically expected, the boxplot in figure 1.9 clearly illustrates a positive trend which individuals in higher income brackets generally allocate

significantly larger budgets for their trips compared to those in lower income brackets. This underscores the importance of income level as a primary factor influencing travel spending.

ANOVA Results – Travel Budget by Booking Advance

```
Table 1.6: ANOVA Results - Travel Budget by Booking Advance
====================================================================

Descriptive Statistics by Group:
--------------------------------------------------------------------
3-6 months           N=104 Mean= 4683.55 SD= 2102.80 Range=1304.00- 9526.00
Within 1 week        N= 14 Mean= 6525.36 SD= 1459.93 Range=3488.00- 9218.00
1-4 weeks            N= 59 Mean= 4221.51 SD= 1597.97 Range=1302.00- 8559.00
More than 6 months   N= 73 Mean= 4393.95 SD= 1954.52 Range=1148.00-10823.00


ANOVA Table:
--------------------------------------------------------------------
        Source           SS  df          MS      F p-value
Between Groups  64163563.74   3 21387854.58 5.8054  0.0008
 Within Groups 906304041.50 246  3684162.77      -       -
         Total 970467605.24 249           -      -       -
```

Table 1.6 reveals a statistically significant difference in budgets based on booking lead time (p = 0.0008). Figure 1.14 suggests a pattern that travelers who book 'Within 1 week' in advance tend to have significantly higher budgets compared to those who book further in advance to their departure date.

ANOVA Results – Travel Budget by Number of Trips Per Year

```
Table 1.7: ANOVA Results - Travel Budget by Number of Trips Per Year
====================================================================

Descriptive Statistics by Group:
--------------------------------------------------------------------
6 times or more      N= 13 Mean= 5311.15 SD= 2352.05 Range=1628.00- 9178.00
1-2 time(s)          N= 97 Mean= 4651.86 SD= 2146.81 Range=1282.00- 9736.00
3-5 times            N= 83 Mean= 4822.48 SD= 1854.99 Range=1148.00-10823.00
0                    N= 57 Mean= 3995.26 SD= 1624.04 Range=1231.00- 8477.00


ANOVA Table:
--------------------------------------------------------------------
        Source           SS  df          MS      F p-value
Between Groups  31777123.79   3 10592374.60 2.7759  0.0419
 Within Groups 938690481.45 246  3815814.97      -       -
         Total 970467605.24 249           -      -       -
```

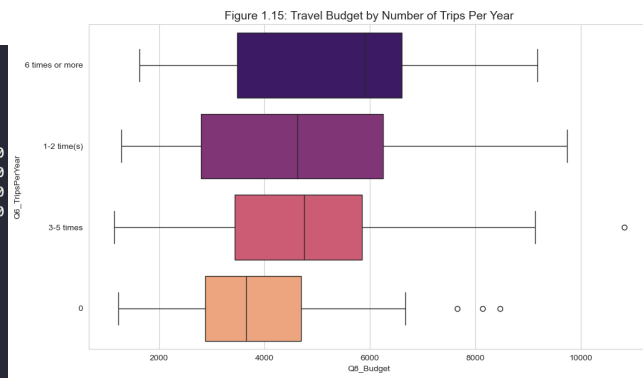


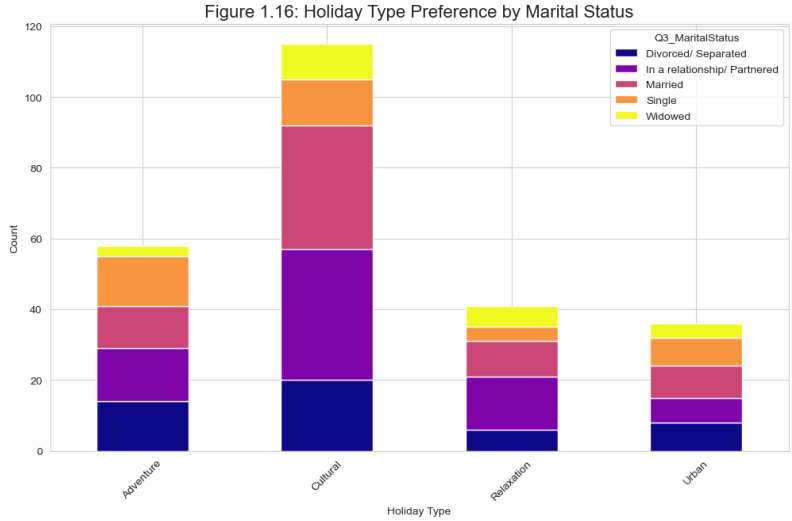
Table 1.7 and figure 1.15 show that individuals who travel less frequently tend to allocate a lower budget for each individual trip.

Chi-Squared Test – Association between Holiday Type and Marital Status

Purpose: To test for a statistically significant association between a traveler's preferred holiday type and their marital status.

```
--- Table 1.8: Chi-squared Test Results - Holiday Type by Marital Status ---
======================================================================
       Test Statistic   Value
Chi-squared statistic 14.0188
              P-value  0.2995
   Degrees of freedom      12
          Sample size     250
======================================================================
```

Figure 1.16: Holiday Type Preference by Marital Status



According to table 1.8 and figure 1.16, the Chi-squared test resulted in a p-value of 0.2995, which is greater than the significance level of 0.05. Therefore, we fail to find a statistically significant association between holiday type preference and marital status in this sample. While the stacked bar chart from figure 1.16 shows some variations in the proportions, these differences are not statistically strong enough to conclude a definitive relationship in the broader population based on this data. Marital status alone may not be a primary driver of holiday type choice within this group.

Machine Learning Analysis

Purpose: To leverage machine learning techniques to gain deeper insights into traveler preferences, specifically focusing on identifying the key factors that predict a preference for adventure travel. This addresses the goal understanding market requirements to inform the development of targeted travel packages. A supervised learning approach was adopted to directly predict a specific, high-value traveler preference.

The Exploratory Data Analysis highlighted 'Adventure' as one of the most popular holiday types, yet also one where preferences are polarized (high interest vs. low interest).

Understanding who prefers adventure travel is crucial for the agency to effectively design and market adventure-specific packages versus relaxation or cultural packages.
To facilitate this prediction task, the original multi-category 'Q7_HolidayType' variable was transformed into a binary target variable named 'Target_AdventurePreference'. Respondents who chose 'Adventure' were coded as 1 (Prefers Adventure), and all other respondents (choosing other types) were coded as 0 (Does Not Prefer Adventure). This simplification allows us to build powerful binary classification models focused on this key market differentiator.

Rationale for Model Selection

Four distinct classification models were chosen to provide a comprehensive comparison and identify the most effective approach for predicting adventure preference with this dataset.

- **Logistic Regression:** It is selected as a robust and interpretable baseline model. It helps identify linear relationship between the predictors (age, budget, annual household income, etc.) and the likelihood of preferring adventure travel.
- **Decision Tree:** The tree model is chosen for its high interpretability. The model creates a tree-like structure of decision rules (e.g., 'if age < 35 and budget <$4,000 then Prefers Adventure') which can be easily understood and translated into business logic.
- **Naïve Bayes:** A probabilistic model is selected for its efficiency and ability to perform well even with potential correlations between features despite its naïve assumption of independence which can sometimes limit accuracy.
- **Random Forest:** An advanced ensemble method is chosen for its potential high accuracy and robustness. By combining many decision trees, it often captures complex patterns and interactions between features that simpler models might miss, leading to superior predictive performance.

Machine Learning Model Performance Comparison

```
----------------- Table 1.9: Model Comparison (Ranked by Test Accuracy) -----------------
              Model  Mean Train Accuracy  Train Std  Mean Test Accuracy  Test Std  Overfitting Gap
      Random Forest                0.795      0.015               0.684     0.058            0.111
      Decision Tree                1.000      0.000               0.636     0.117            0.364
Logistic Regression                0.668      0.023               0.564     0.095            0.104
        Naive Bayes                0.286      0.025               0.268     0.044            0.018
```

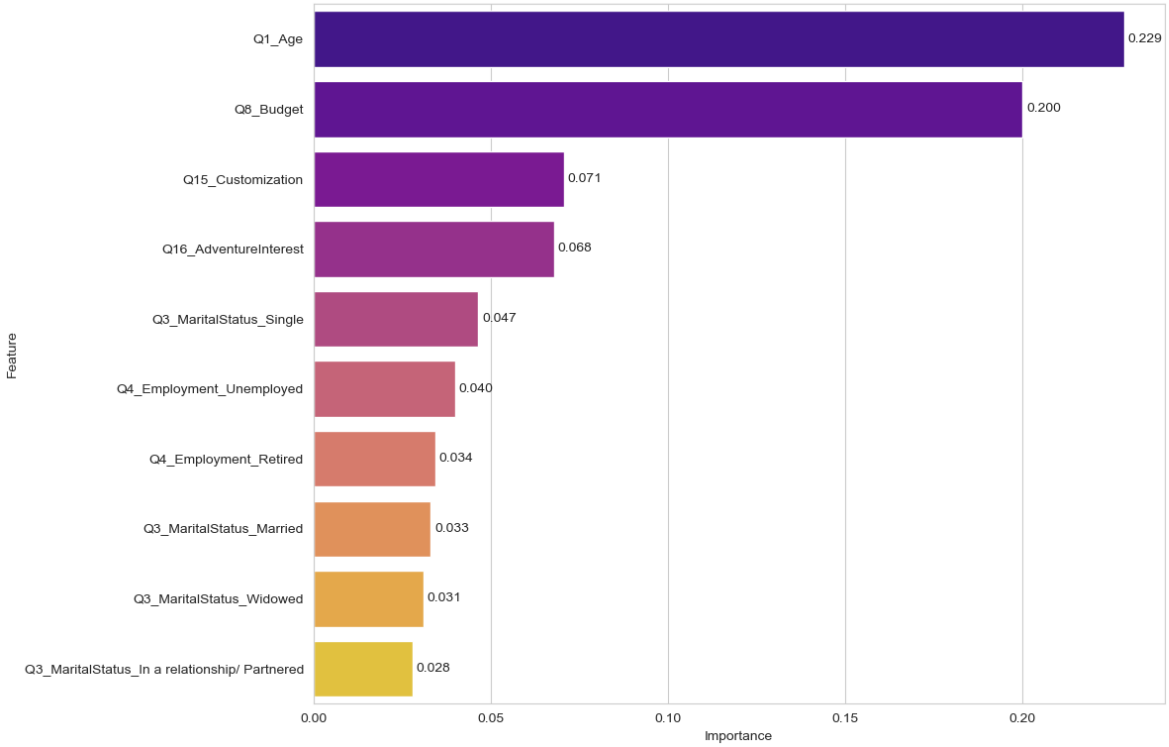
Figure 1.17: Model Performance Comparison

A 10-fold cross-validation approach was used to obtain reliable estimates of each model's performance on unseen data. Accuracy was chosen as the primary evaluation metric.

According from the results, the Random Forest model emerged as the top performer, achieving the highest mean accuracy of approximately 68.4%. The Decision Tree performs poorly because of its huge overfitting gap with 63.6% of accuracy with Logistic Regression and Naïve Bayes lagged behind. The Random Forest's superior accuracy suggests that predicting adventure preference likely involves complex interactions between traveler characteristics, which the ensemble nature of this model is well-suited to capture.

Top 10 Feature Importances for Predicting Adventure Preference

Figure 1.18: Top 10 Feature Importances for Predicting Adventure Travel Preference (Random Forest)



Purpose: Having identified Random Forest as the most accurate predictor, this analysis aims to understand which traveler characteristics the model found most influential when determining if someone prefers adventure travel.

The Random Forest model was trained one final time on the entire dataset. Feature importances, which quantify the contribution of each predictor variable to the model's accuracy, were then extracted.

Figure 1.18 provides actionable insights for the travel agency by highlighting the key drivers of adventure preference. The three most important factors identified by the model are:

- Age (Q1_Age): With an importance score of 0.229, age is the single most significant predictor. This strongly confirms the common understanding that younger travelers are generally more inclined towards adventure-style holidays compared to older travelers.
- Budget (Q8_Budget): Budget is the second most crucial factor, with an importance score of 0.2, highlighting that financial capacity plays a major role in choosing adventure travel, likely distinguishing between different types of adventure seekers (e.g., budget backpackers vs. those undertaking more expensive expeditions).
- Customization Importance (Q15_Customization): The desire for trip customization ranks third, with a score of 0.071. This points out that individuals who prefer adventure travel also tend to value having more control over planning their itinerary and activities, aligning with the idea of seeking unique or personalized experiences.

These top three findings strongly suggest that marketing efforts for adventure travel should primarily focus on younger demographics, while also segmenting offers based on budget level and emphasizing flexible or customizable trip options.

Illicit Drug Use Among Young People: Policy Guidelines

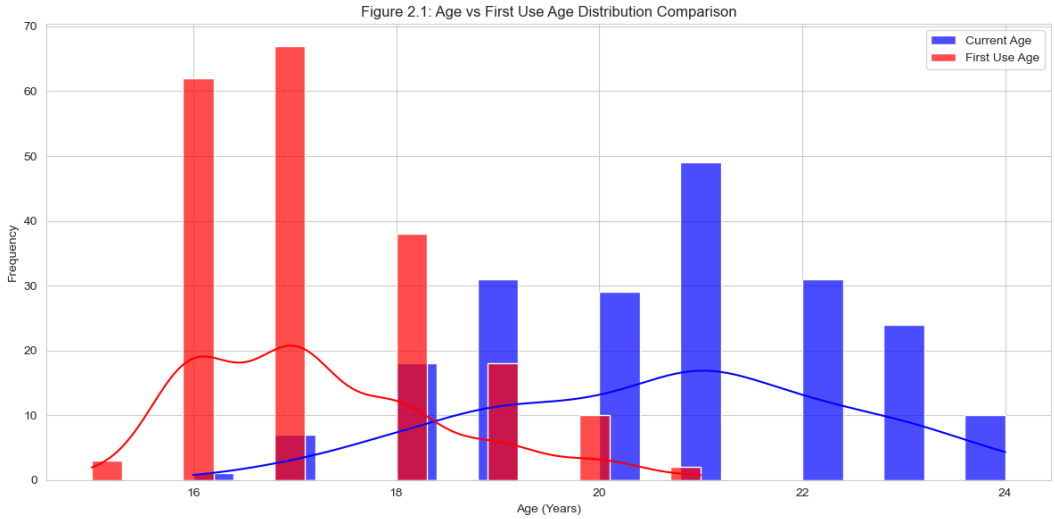
Exploratory Data Analysis

Purpose: To summarize the core characteristics of the 200 respondents, demonstrating a foundational understanding of the sample's demographics, behaviors, and attitudes before conducting inferential testing.

Numerical Variables

```
--- Table 2.1: Descriptive Statistics for Numerical Columns ---
       Q1_Age  Q15_FirstUseAge  Q16_WeeklySpend
count  200.00           200.00           200.00
mean    20.66            17.22            42.74
std      1.80             1.22            24.63
min     16.00            15.00             9.45
25%     19.00            16.00            23.47
50%     21.00            17.00            36.71
75%     22.00            18.00            57.38
max     24.00            21.00           116.71
```

Age vs First Use Age Distribution Comparison



According to table 2.1 and figure 2.1, the age distribution of the respondents is concentrated in the early twenties, with a mean age of approximately 20.66 years which are in young generation. Moreover, the age of first use is similarly clustered in the late teens (around 16-19 years), suggesting that initiation into drug use typically occurs during secondary school or shortly after.

Weekly Spending Distribution on Drugs

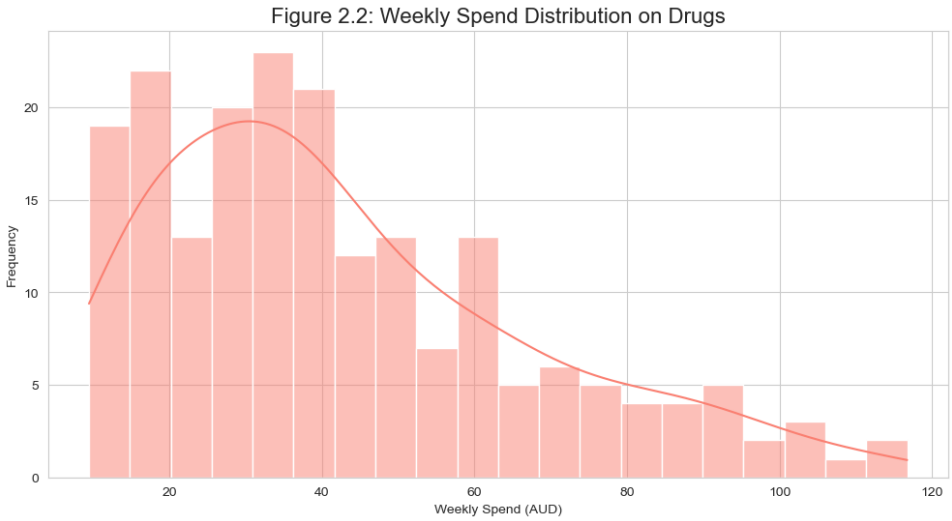


Figure 2.2 points out that the distribution of weekly spending on drugs is right-skewed, which means that while most users spend a modest amount (under $45/week), a smaller subset of individuals spends significantly more, with some exceeding $90/week. The average weekly spend across the sample is approximately $43.

Key Categorical Variables

Distribution of Gender

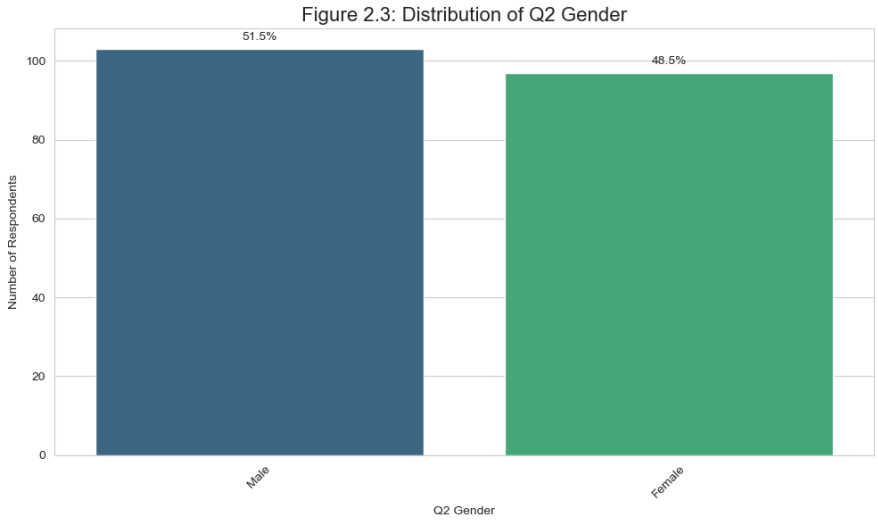
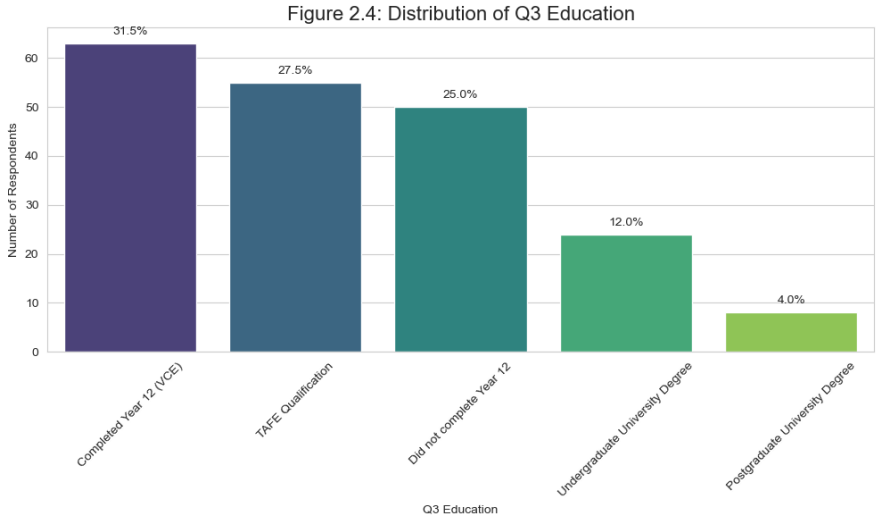


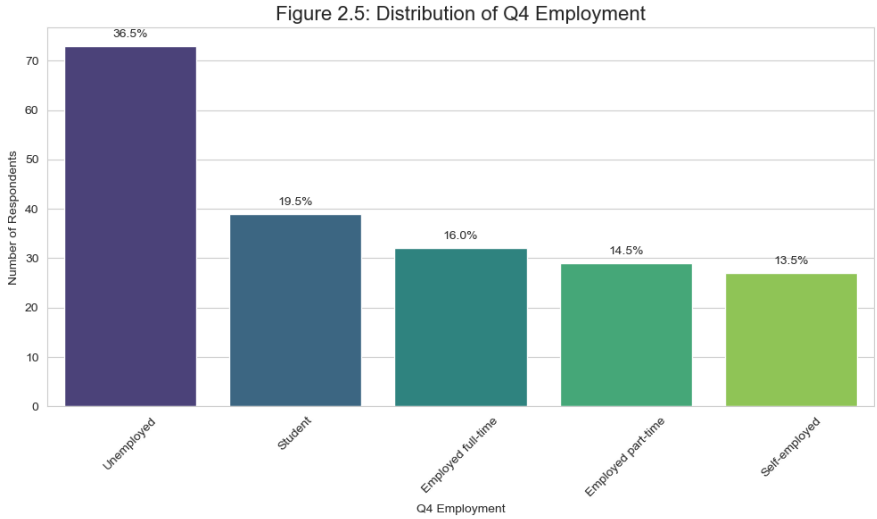
Figure 2.3 shows a near-even split between genders.

Distribution of Education Levels



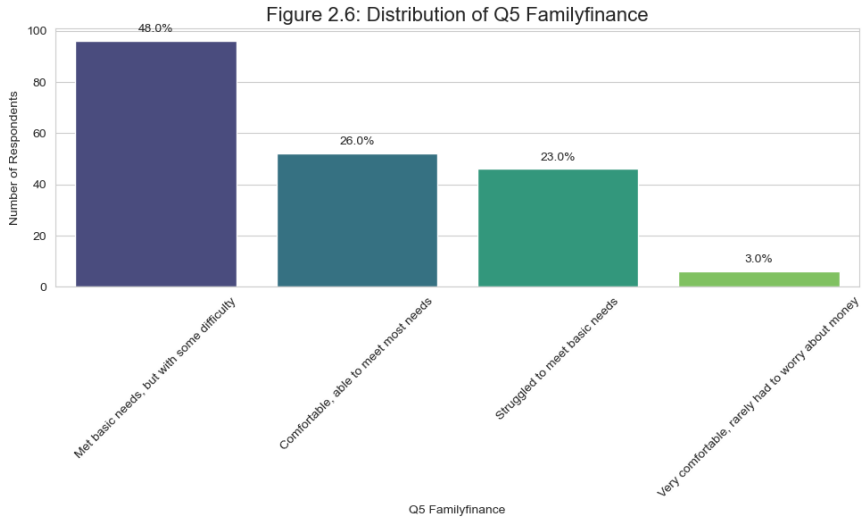
The majority of respondents have completed at least Year 12, followed by TAFE Qualification. Only a few of them are holding or pursuing a university degree.

Distribution of Employment Status



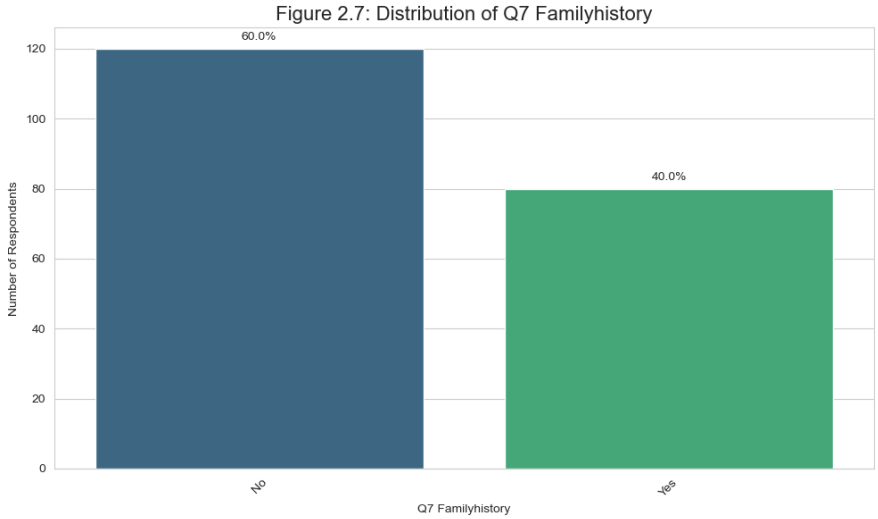
The employment status is varied with ‘Student’ and ‘Unemployed’ as the most common categories, reflecting the young age of the sample from figure 2.5.

Distribution of Family Finance



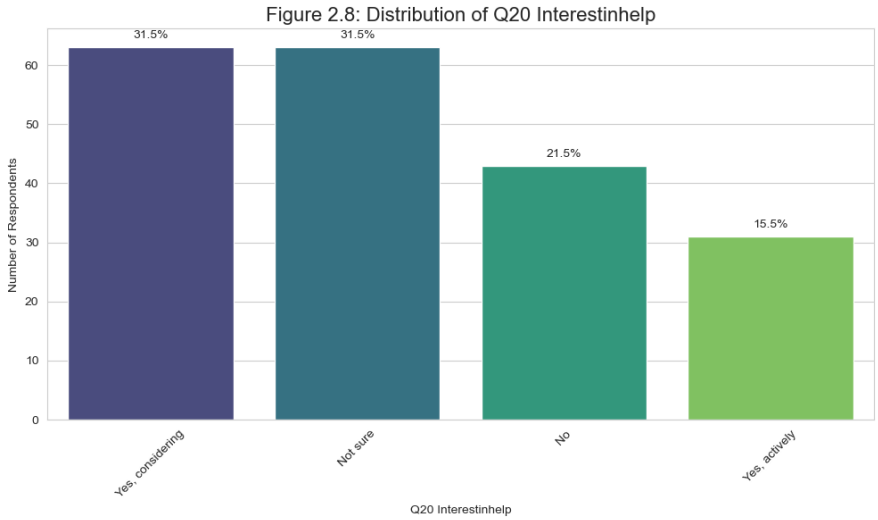
In terms of family financial background, most respondents grew up in households that were 'Comfortable' or 'Met basic needs', indicating that drug use in this sample might not be exclusively linked to severe economic deprivation.

Distribution of Family History on Drug and Alcohol



A majority of respondents report having a family history of drug or alcohol addiction, suggesting a potential intergenerational link from figure 2.7.

Distribution of Interest in Help



From figure 2.8, there is a positive outlook for intervention, as a clear majority of respondents indicated they are either 'actively seeking help' or would 'consider' it. This represents a crucial opportunity for public health initiatives.

Distribution of Psychological and Social Factors

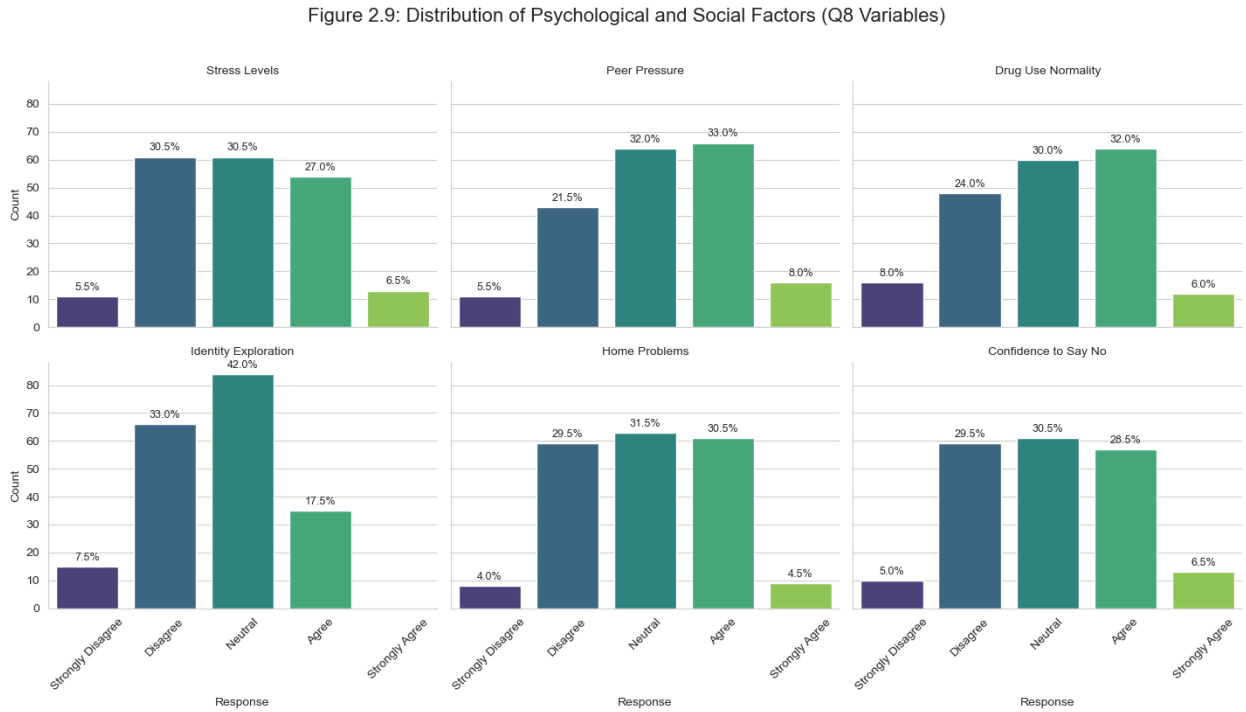
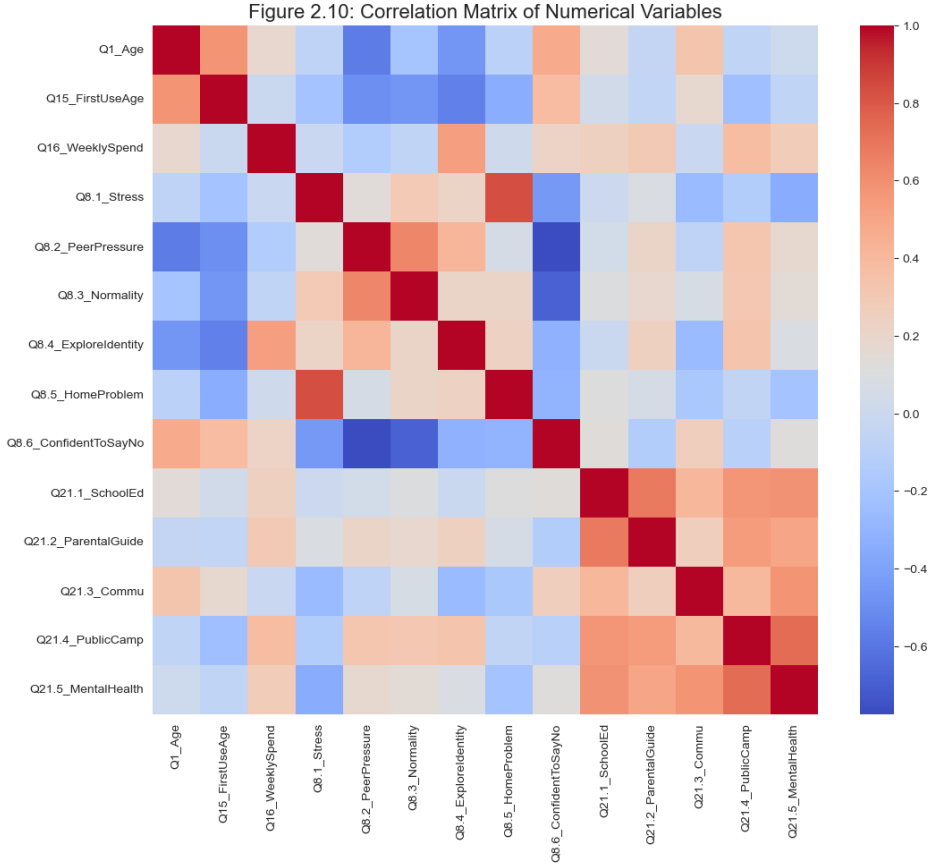


Figure 2.9 points out that on average, respondents tended to agree that they use drugs to cope with stress and home problems. In contrast, respondents tended to strongly disagree with statements related to external social factors. The distribution of peer pressure is heavily skewed to the left ('Disagree'), suggesting that the respondents' drug use is not primarily driven by a need to fit in with their social circle. Similarly, there was general disagreement with the idea that drug use is a way to 'explore identity'. For the confidence to

say no to drugs, the responses are skewed slightly to the right, suggesting that despite their use, they perceive themselves as being in control of their decisions, viewing their drug use as a conscious choice rather than a result of being unable to refuse.

Correlation Matrix of Numerical Variables

Figure 2.10: Correlation Matrix of Numerical Variables



The correlation matrix from figure 2.10 visualizes the relationships between all numerical variables. It shows that there is a strong positive correlation between using drugs to cope with stress (Q8.1_Stress) and using them to cope with home problems (Q8.5_HomeProblem). Both of these factors also show a moderately positive correlation with weekly spending (Q16_WeeklySpend), suggesting that individuals who use drugs as a coping mechanism tend to have higher drug-spending habits.

Another insight is the negative correlation between peer pressure (Q8.2_PeerPressure) and weekly spending (Q16_WeeklySpend) which suggests that higher perceived peer pressure is associated with lower spending. This may be because heavy, high-spending users are driven by internal factors like addiction, while lower-spending, more social users are the ones who feels more external peer pressure.

There is a slight positive correlation between current age (Q1_Age) and weekly spending (Q16_WeeklySpend), suggesting that as users get older within this young-adult sample, their spending on drugs tends to increase. This could be due to factors like increased

tolerance, a long history of use, or greater disposable income compared to their teenage counterparts.

Foundational Inferential Analysis

Purpose: To conduct inferential statistical tests to draw conclusions about the wider population, including estimating the proportion of users who are open to receiving help and testing for significant differences in drug spending habits across key demographic groups.

95% CI for Interest in Help

```
--- Table 2.2: Proportion of Respondents Interested in Seeking Help ---
==========================================================================
                    Metric  Value
      Sample Size (Total)    200
Number Interested in Help     94
        Sample Proportion 0.4700
       95% CI Lower Bound 0.4020
       95% CI Upper Bound 0.5391
==========================================================================
```

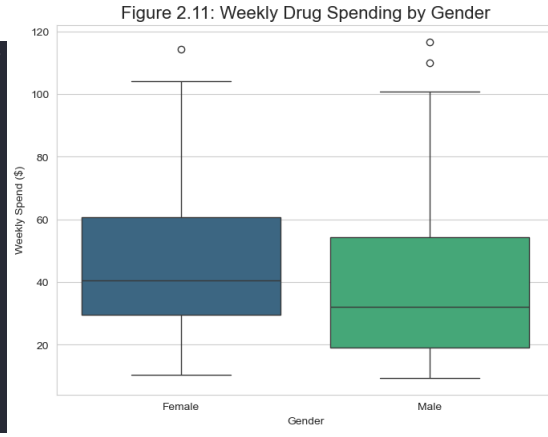
Table 2.2 shows that 47% of the sample expressed an interest in receiving help. We can be 95% confident that the true proportion of all 2,000 respondents in the target population who are interested in help lies between 40.20% and 53.91% which are receptive to intervention and should be targeted by the public health outreach.

T-test – Weekly Drug Spending by Gender

```
--- Table 2.3: T-Test Summary for Weekly Spending by Gender ---
T-Test Results: Weekly Spending by Gender
==============================================
                Statistic   Value
              T-statistic -1.3057
                  P-value  0.1932
        Degrees of Freedom   195.7
        Sample Size (Male)     103
      Sample Size (Female)      97
  Mean Weekly Spend (Male)  $40.55
Mean Weekly Spend (Female)  $45.07
    SD Weekly Spend (Male)  $26.50
  SD Weekly Spend (Female)  $22.38

==============================================
```

Figure 2.11: Weekly Drug Spending by Gender



The t-test results shown in Table 2.3 find no statistically significant difference in mean weekly spending between males ($40.55) and females ($45.07), with a p-value of 0.1932, suggesting that gender is not a significant factor in determining the amount of money

spend on drugs withing this sample. Figure 2.11 also support this, showing highly overlapping distributions between two groups.

T-test – Weekly Drug Spending by Family History

```
--- Table 2.4: T-Test Summary for Weekly Spending by Family History ---
T-Test Results: Weekly Spending by Family History
==================================================
              Statistic   Value
            T-statistic -0.6749
                P-value  0.5006
     Degrees of Freedom   174.3
      Sample Size (Yes)      80
       Sample Size (No)     120
Mean Weekly Spend (Yes)  $41.31
 Mean Weekly Spend (No)  $43.69
  SD Weekly Spend (Yes)  $24.03
   SD Weekly Spend (No)  $25.07

==================================================
```

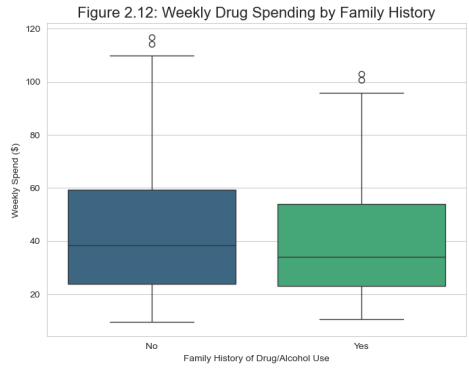


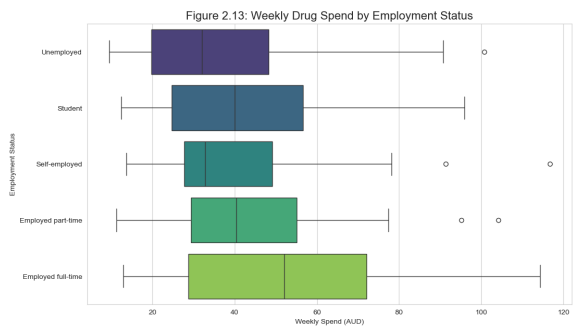
Table 2.4 reveals no statistically significant difference in mean weekly drug spending based on family history with p-value 0.5006, suggesting that a family history of drug/ alcohol addiction is not a contributing factor to drug use habits.

ANOVA Test – Weekly Drug Spending by Employment Status

```
Table 2.5: ANOVA Table - Weekly Drug Spending by Employment Status
=======================================================

Descriptive Statistics by Employment Status:
-------------------------------------------------------
Unemployed           N= 73 Mean=$ 37.17 SD=$ 21.95 Range=$ 9.45-$100.78
Student              N= 39 Mean=$ 43.06 SD=$ 23.86 Range=$12.32-$ 95.88
Self-employed        N= 27 Mean=$ 42.75 SD=$ 24.61 Range=$13.63-$116.71
Employed part-time   N= 29 Mean=$ 43.64 SD=$ 23.41 Range=$11.16-$104.17
Employed full-time   N= 32 Mean=$ 54.22 SD=$ 29.39 Range=$12.89-$114.33


ANOVA Results:
-------------------------------------------------------
        Source        SS  df      MS      F p-value
Between Groups   6509.35   4 1627.34 2.7789  0.0281
 Within Groups 114192.99 195  585.61      -       -
         Total 120702.34 199       -      -       -
```



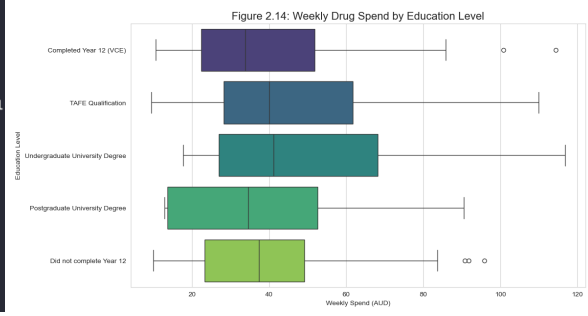
The ANOVA test from table 2.5 reveals a statistically significant result (p-value =0.0281), indicating that there is a significant difference in the average amount of money spent on drugs between at least two of the employment groups. Figure 2.13 suggests that 'Unemployed' respondents may have a different spending pattern compared to 'Students' or the 'Employed'.

ANOVA Test – Weekly Drug Spending by Education Level

```
Table 2.6: ANOVA Table – Weekly Drug Spending by Education Level
==========================================================

Descriptive Statistics by Education Level:
----------------------------------------------------------
Completed Year 12 (VCE) N= 63 Mean=$ 40.07 SD=$ 23.58 Range=$10.55–$114.33
TAFE Qualification   N= 55 Mean=$ 46.91 SD=$ 26.85 Range=$ 9.45–$109.97
Undergraduate University Degree N= 24 Mean=$ 48.26 SD=$ 26.40 Range=$17.64–$116.71
Postgraduate University Degree N=  8 Mean=$ 38.44 SD=$ 27.22 Range=$12.89–$ 90.54
Did not complete Year 12 N= 50 Mean=$ 39.54 SD=$ 21.79 Range=$ 9.96–$ 95.88


ANOVA Results:
----------------------------------------------------------
        Source        SS  df     MS      F p-value
Between Groups   2798.56   4 699.64 1.1571  0.3311
 Within Groups 117903.78 195 604.63      –       –
         Total 120702.34 199      –      –       –
```



According to the ANOVA test result in table 2.6, it is not statistically significant (p-value = 0.3311) so we cannot conclude that there is a significant difference in mean weekly drug spending based on the respondents' level of education, aligned with the overlap between groups shown in Figure 2.14.

Machine Learning Analysis

Purpose: To build a predictive model to identify which individuals are most likely to be receptive to intervention. The goal is to create a tool that can help public health officials target their outreach more effectively.

To achieve this, several machine learning classification models were trained to predict the binary outcome, whether a person is interested in receiving help or not.

Rationale for Model Selection

A diverse suite of four classification models was chosen to ensure a comprehensive and robust comparison. Each model approaches the prediction task differently, allowing us to identify the most suitable technique for this specific problem.

- **Logistic Regression:** It was chosen as a strong baseline model. It works by finding a linear relationship between the predictor variables and the probability of a person being interested in help. It's highly interpretable and provided a good benchmark for more complex models.
- **Decision Tree:** This model partitions the data based on the most informative features at each step, creating a tree of decision rules. It was selected for its high interpretability, as the resulting 'rules' are easy to understand and explain.
- **Naïve Bayes:** It calculates the probability that an individual belongs to the 'interested in help' class, given their specific set of features. It's computationally efficient and performs well even with a smaller number of features.
- **Random Forest:** It is a more advanced 'ensemble' model because instead of building one single Decision Tree, it builds hundreds of them, each on a slightly different subset of the data and features. To make a final prediction, it takes a majority vote from all the individual trees. It was chosen because its 'wisdom of the

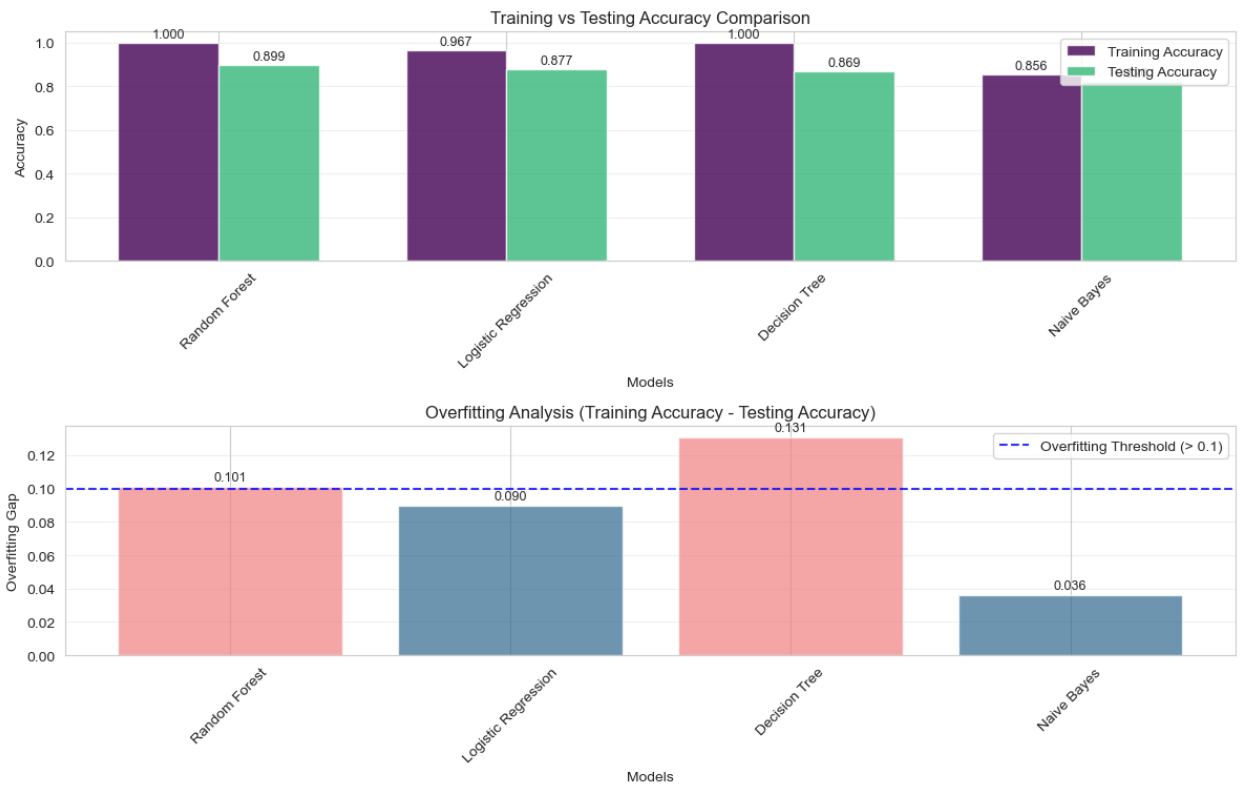
crowd' approach often leads to higher accuracy and better generalization, making it one of the most powerful and popular classification algorithms.

Machine Learning Model Performance Comparison

A robust 10-fold cross-validation technique was employed to evaluate the performance of each model. The final accuracy is the average of 10 trials which provides a highly reliable measure of each model's true performance. The criterion of overfitting is set at 0.1 which is quite conservative threshold.

```
------------------ Table 2.7: Model Comparison (Ranked by Test Accuracy) ------------------
               Model  Mean Train Accuracy  Train Std  Mean Test Accuracy  Test Std  Overfitting Gap
       Random Forest                1.000      0.000               0.899     0.080            0.101
 Logistic Regression                0.967      0.008               0.877     0.085            0.090
       Decision Tree                1.000      0.000               0.869     0.079            0.131
         Naive Bayes                0.856      0.013               0.820     0.125            0.036
```

Figure 2.15: Model Performance Comparison



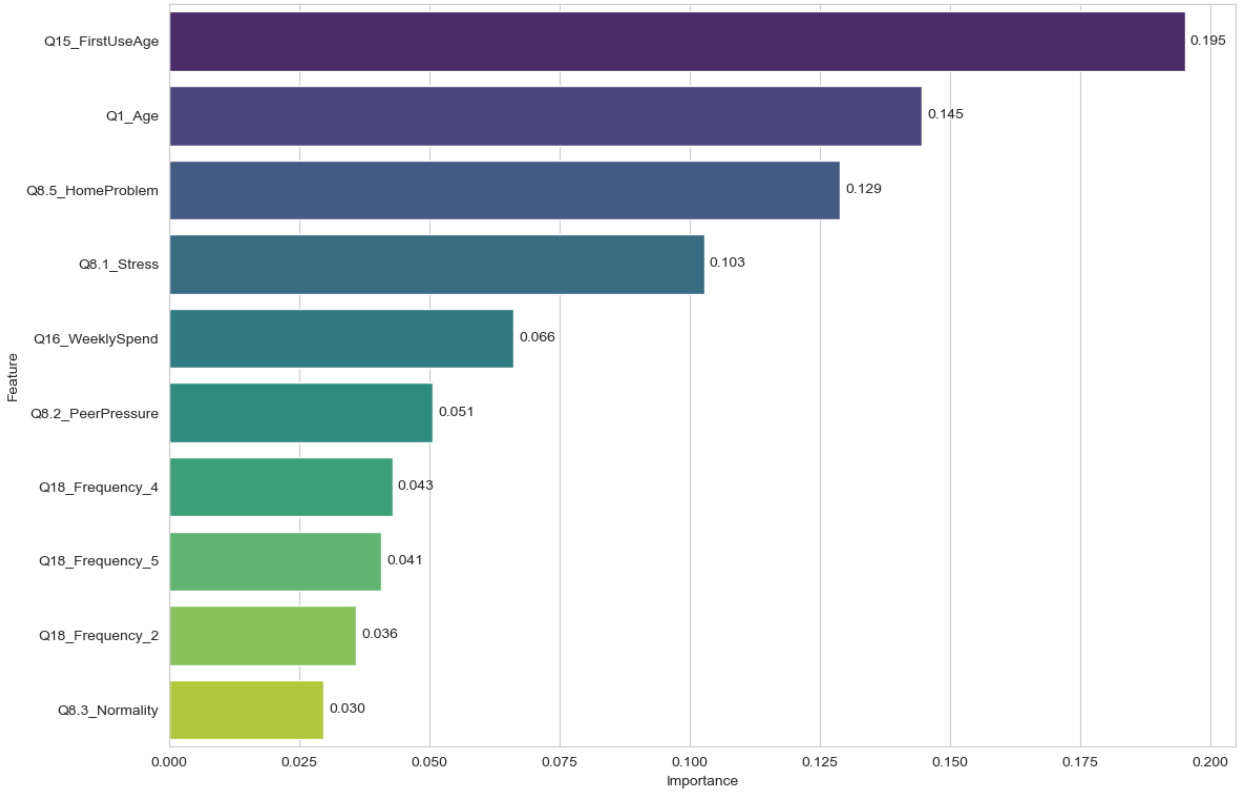
According from the result shown in Table 2.7 and Figure 2.15, The Random Forest Model achieved the highest mean accuracy of approximately 89.9%, significantly outperforming the other models. The Logistic Regression and Decision Tree models performed respectably (around 87.7% and 86.9%), while the Naïve Bayes model was the least accurate (around 82.0%), representing as the weakest performer, likely due to its core naïve assumption that all features are independent. This assumption is not true in this dataset. For example, according to figure 2.10, stress and home problems are positively correlated, age and weekly spending on drugs are slightly positively correlated, and peer pressure and weekly spending on drugs are negatively correlated.

On the other hand, the superior performance of the Random Forest is likely because its ensemble nature, which combines the predictions of many diverse decision trees, allows it to capture more complex, non-linear patterns in the data and makes it less prone to overfitting compared to a single Decision Tree.

All models potentially performed well without massive overfitting, given the largest gap between training accuracy and testing accuracy at 0.131 from Decision Tree, it is still considered as low overfitting problem.

Top 10 Feature Importances for Predicting Interest in Help

Figure 2.16: Top 10 Feature Importances for Predicting Interest in Help



Purpose: Having identified the Random Forest as the most accurate model, the next step is to look inside the model to identify which features it found most important when making it predictions.

The best-performing model (Random Forest) was trained on final time on the entire dataset and extracted its 'feature importances', which is a score the model assigns to each variable based on how much it contributed to improving the accuracy of its predictions. A higher score means the feature was more influential.

The feature importance analysis reveals the profile of an individual who is most likely to be receptive to help. The most critical predictors are not simple demographics but are instead a mix of behavioral and psychological factors:

- Weekly Spending on Drug (Q16_WeeklySpend): This was the single most important factor. This result strongly suggests that individuals who have a more severe (and expensive) drug habit are more likely to recognize the need for help from this sample set.
- Coping with Home Problem and Stress (Q8.5_HomeProblem and Q8.1_Stress): This indicates that users who explicitly stat they use drugs to cope with domestic issues and stress are a key group who may be open to intervention.
- Age (Q1_Age and Q15_FirstUseAge): Both of age-type responses are significant predictors, suggesting that an individual's life stage and the duration or their drug use play a crucial role in their willing ness to seek help.